Research statement: A statistical study of topological and geometric descriptors

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Topological Data Analysis (or TDA) is a set of techniques used to extract topological and geometric information from complex datasets, think of graphs, time series, or point clouds close to manifolds for instance. During my Ph.D. thesis, I explored two different approaches used in this field from a statistical point of view, namely manifold inference and persistent homology theory.

Minimax estimation in manifold inference

The first class of problems I have studied broadly falls under the term of "manifold inference". Such problems all make the key assumption that some observed dataset $\mathbb{X}_n = \{X_1, \dots, X_n\}$, which lies in a possibly high-dimensional space $\mathbb{R}^D$, is actually close to some low-dimensional shape M (which is modeled as a d-dimensional manifold with $d < D$). This manifold assumption generalizes common hypotheses made in high-dimensional statistics, such as sparsity of the dataset, the single (or multi) index model [LZZL13], or the presence of a low-dimensional linear subspace which fits the dataset. Note that, unlike sparsity, being close to a manifold is invariant by isometry, and is also invariant by smooth reparametrizations of the coordinates.

A program which has been fruitfully developed over the past decade consists in using $\mathbb{X}_n$ to design a variety of descriptors that help to understand the geometry of the underlying shape M, such as its dimension [KRW19], its homology groups [NSW08], its curvature [AKC+19], etc. From a statistical perspective, the question of the optimality of those estimators is then raised, where optimality has to be understood in a minimax sense: if $\mathcal{P}$ is a set of probability distributions on $\mathbb{R}^D$, and $\theta : \mathcal{P} \to (E, d)$ is a functional of interest (taking its values in some metric space (E, d)), then the minimax risk of estimation of θ on the statistical model $\mathcal{P}$ given n observations is equal to

$$\mathcal{R}_n(\theta; \mathcal{P}; d) := \inf_{\hat{\theta}_n} \sup_{\mu \in \mathcal{P}} \mathbb{E}_{\mu^{\otimes n}}[d(\hat{\theta}_n, \theta(\mu))], \tag{1}$$

where the infimum is taken over all estimators $\hat{\theta}_n : (\mathbb{R}^D)^n \to E$.

Adaptive manifold estimation A first issue consists in designing statistical models that encapsulate the regularity of the underlying manifold. The key quantity to do so is the reach of a manifold $\tau(M)$, given by the distance between the manifold and its medial axis. Introduced in [Fed59], it captures both the local curvature of M and its global regularity. The model $\mathcal{P}_{d,\tau_{\min},f_{\min},f_{\max}}$ is then the set of all laws μ supported on a d-dimensional manifold M with $\tau(M) \geq \tau_{\min}$, having a density f on M with $0 < f_{\min} \leq f \leq f_{\max}$.

If we consider the problem of the reconstruction of the manifold M for the Hausdorff distance d_H, the minimax risk on those models is known to be of order (see [GPPV$^+$12, KZ$^+$15])

$$\mathcal{R}_n(M; \mathcal{P}_{d,\tau_{\min},f_{\min},f_{\max}}, d_H) \asymp \left(\frac{\ln n}{n}\right)^{2/d}, \tag{2}$$

while several minimax estimators have been proposed in the literature [AL18, GPPV$^+$12, PS19]. Those estimators all depend on several hyperparameters, which, to be minimax, have to be properly tuned with respect to the parameters $f_{\min}, f_{\max}, \tau_{\min}$ defining the model. In practice, those parameters are unknown, and the tuning of the hyperparameters turns out to be a real issue.

The problem of the tuning of hyperparameters is not limited to the framework of manifold inference and is a classical issue in statistics. A standard way to overcome it consists in designing *adaptive* estimators, i.e. estimators which are simultaneously minimax on a large class of models. In a work submitted to the Electronic Journal of Statistics [Div20], I showed that it is possible to design such an adaptive estimator for the problem of manifold estimation. This was made possible by introducing a particularly simple manifold estimator $\hat{M}_t$, depending on a single index t, and defined as the union of simplexes on $\mathbb{X}_n$ of size roughly equal to t. After showing that this estimator is minimax for a certain choice t_{opt} (depending on $d, \tau_{\min}, f_{\min}, f_{\max}$), I developed a procedure to select a scale t almost as good as the optimal one. This is done by computing the so-called convexity defect function of $\mathbb{X}_n$ [ALS13], defined as $t \mapsto d_H(\hat{M}_t, \mathbb{X}_n)$. I have shown that this function exhibits a sharp change of behavior around a threshold value $t^*(\mathbb{X}_n)$, and that selecting a value t just above this threshold gives an estimator with almost the same risk as $\hat{M}_{t_{opt}}$, leading to an adaptive estimator on the class of models $\mathcal{P}_{d,\tau_{\min},f_{\min},f_{\max}}$ for $\tau_{\min}, f_{\min}, f_{\max} > 0$.

Wasserstein estimation on manifolds A second question I considered is the reconstruction of the law μ generating the observations itself. It is possible to show that such a reconstruction problem is degenerate from a minimax perspective for a large class of standard losses on the set of probability measures (total variation, Hellinger distance, or other f-divergences) while being tractable for distances metricizing the weak convergence of measures, such as Wasserstein distances W_r. Even in the case of densities on the cube $[-1,1]^d$, the study of the reconstruction of smooth measures in terms of Wasserstein losses is very recent, having only been considered so far in [WB19b], where minimax rates are exhibited on models taking into account the regularity of some underlying density on the cube $[-1,1]^d$. Building upon recent results from [AL$^+$19] on the reconstruction of local parametrizations of $\mathcal{C}^k$ manifolds, I showed that it is possible to adapt classical kernel density estimators to the case of manifold-supported observations $\mathbb{X}_n$, while retaining the same minimax rates of convergence as for densities on the cube. This topic is the purpose of a current work in progress.

Understanding properties of random persistence diagrams

Persistent homology [EH10] is a popular approach in TDA, which provides efficient mathematical and algorithmic tools to understand the topology of some dataset (e.g. a point cloud or a time-series) by tracking the evolution of its homology at different scales. It consists in an algebraic machinery which takes as an input an increasing sequence of topological spaces $(X(t))_{t\geq 0}$ (think of the sublevel sets $X(t) = f^{-1}(-\infty, t]$ of a function $f : E \to \mathbb{R}$) and outputs a descriptor called a *persistence diagram*. The persistence diagram consists of a multiset a in the upper half-plane $\Omega := \{x = (t_1, t_2) \in \mathbb{R}^2,\ t_1 < t_2\}$, where each point (t_1, t_2) of the multiset a accounts for a topological feature (connected component, hole, etc.) which appeared in the filtration at the scale t_1 and disappeared at the scale t_2. Equivalently, a can

be seen as a locally finite discrete measure $\sum_{x\in a}\delta_x$ on Ω. As an example, one can consider f to be the distance function from some set A: the corresponding diagram, called the Čech persistence diagram of A, summarizes the topology of the set A in a multiscale fashion.

Let $\mathcal{D}$ be the set of persistence diagrams. The proximity between two persistence diagrams $a, b \in \mathcal{D}$ is usually measured thanks to d_p distances, defined as follows. Let $\Gamma(a,b)$ be the set of bijections between $a \cup \partial\Omega$ and $b \cup \partial\Omega$, where $\partial\Omega = \{(t,t),\ t \in \mathbb{R}\}$ is the diagonal. Then,

$$d_p(a,b) := \inf_{\gamma\in\Gamma(a,b)} \left(\sum_{x\in a\cup\partial\Omega} |x-\gamma(x)|^p \right)^{1/p}, \tag{3}$$

with usual modification if $p=\infty$ [EH10, Chapter VIII]. The utilization of such distances is motivated by stability results: given two close objects (e.g. two sets close to each other for the Hausdorff distance), the corresponding persistence diagrams are close to each other for the d_p distance in a wide variety of settings.

The geometry of the space of the persistence diagrams The corresponding metric space $(\mathcal{D}_p, d_p)$—where $\mathcal{D}_p = \{a \in \mathcal{D},\ d_p(a,0) < \infty\}$—turns out to be highly non-linear, and deserves further examination. In a work made in collaboration with Théo Lacombe, recently accepted to the Journal of Applied and Computational Topology [DL20], we studied in detail this space by interpreting the distance d_p as a partial optimal transport metric, likewise [FG10]. Taking such an approach has several benefits. First, it allows giving a characterization of convergence with respect to the d_p distances in terms of weak convergence of measures, which provides a very simple way to prove the continuity of applications on the space $\mathcal{D}_p$. Second, making this connection with optimal transport allows us to leverage standard tools from this field (see e.g. [LGL17]). We were in particular able to prove the existence of barycenters for general distributions on the space $\mathcal{D}_p$, generalizing existing results in the literature [MMH11]. Eventually, this optimal transport approach suggests to extend the space $(\mathcal{D}_p, d_p)$ of persistence diagrams to some larger space $(\mathcal{M}_p, \mathrm{OT}_p)$ of so-called persistence measures, i.e. Radon measures on Ω satisfying some finiteness of moments assumption. Such measures appear in different settings of TDA, for instance when studying limit procedures or when computing averages (see below), and the generalization OT_p of the d_p distance makes it possible to compare such objects in a meaningful way.

Representations of persistence diagrams and the choice of the weight function The nonlinear geometry of the metric space $(\mathcal{D}_p, d_p)$ makes the use of standard statistical tools rather awkward. As a workaround, feature maps $\Phi : \mathcal{D}_p \to \mathcal{B}$ (also called representations) are commonly used for the analysis, where $\mathcal{B}$ is a Banach space. A large class of feature maps, which we call linear, depends on some weight functions, the choice of which is a critical issue. In a work with Wolfgang Polonik [DP19], we addressed this choice from two different manners. First, an important criterion to choose a weight function is to ensure the stability of the feature maps with respect to the d_p distances on diagrams. We improved known results on the stability of such maps and extended it to general weight functions. Second, we addressed the choice of the weight function by considering an asymptotic setting; assume that $\mathbb{X}_n$ is an i.i.d. sample from a density on $[0,1]^d$. For the Čech filtration, we characterize the weight functions for which the corresponding feature maps converge as n approaches infinity, and by doing so, we prove laws of large numbers for the total persistences of such diagrams. Those two approaches (stability and convergence) lead to the same simple heuristic for tuning weight functions: if the data lies near a d-dimensional manifold, then a sensible choice of weight function is the distance to the diagonal to the power α with $\alpha \geq d$.

The expected value of a persistence diagram Instead of using feature maps, one could try to perform statistical tasks directly in the space $\mathcal{D}_p$ of persistence diagrams. It is for instance possible to define the expected value of a probability measure P on the space $\mathcal{D}_p$: this is the persistence measure $\mathbb{E}_P[a]$ on Ω such that for all Borel sets $A \subset \Omega$,

$$\mathbb{E}_P[a](A) = \mathbb{E}_P[a(A)], \tag{4}$$

where $a \sim P$ and $a(A)$ is the number of points of the diagram in the region A. In a work made in collaboration with Fréderic Chazal [DC19], we showed that the measure $\mathbb{E}_P[a]$ is a smooth object (i.e. has a smooth density with respect to the Lebesgue measure) in a wide variety of settings, in particular when looking at Čech persistence diagrams of random sets, or at the sublevel sets of the Brownian motion. The regularity of this measure can be leveraged to perform kernel density estimation procedures in the presence of an i.i.d. sample of persistence diagrams of law P, with cross-validation techniques being provably consistent.

Further directions

Minimax estimation of the expected persistence diagram Given i.i.d. observations $a_1, \ldots, a_n$ of some law P on the space of persistence diagrams, one can use the metrics OT_p between persistence measures to study the minimax risk for the estimation of the expected persistence diagram $\mathbb{E}_P[a]$. Building upon convergence results of empirical measures in terms of Wasserstein distances [WB+19a], it is expected that $\frac{1}{n}\sum_{i=1}^n a_i$ converges at rate $n^{-1/(2p)}$ for the OT_p distance. If further regularity is assumed on $\mathbb{E}_P[a]$ (as we know from [DC19] that it is the case in a wide variety of settings), is it possible to find better estimators than the empirical mean?

Oracle inequalities for Wasserstein losses In [WB19b], wavelet estimators are studied with a Wasserstein loss from a minimax perspective. It is shown, that the projection (with respect to some negative Besov norm) of a classical wavelet estimator on the set of densities is minimax for this loss on the class of Besov spaces $B^s_{p,q}$, while a standard thresholding procedure on the wavelet coefficients is shown to lead to an adaptive estimator. It is however not trivial to implement this projection procedure in practice, making the use of those estimators delicate. I conjecture that a kernel density estimator would satisfy the same theoretical properties, without the need for any projection procedure. Also, classical bandwidth selection methods such as the Goldenshluger-Lepski method could lead to oracle inequalities on such a family of estimators.

Spectral clustering and the estimation of the spectrum of the Laplacian on a manifold Spectral clustering methods consist in building an appropriate neighborhood graph on some dataset, and then use the spectral properties of the Laplacian on the graph to partition the dataset into different clusters. It has been shown [TGHS20] that in the case of dataset close to manifolds, such a procedure amounts to estimating the eigenvalues and the eigenvectors of the Laplace-Beltrami operator of the manifold. This raises the question of the optimality of such a procedure: I conjecture that it is possible to leverage the smoothness of the manifold to estimate the spectrum of the Laplace-Beltrami operator with faster rates of convergence. Such an estimation could be made by studying the spectral properties of convolution operators (likewise [LM+10]) on proper $\mathcal{C}^k$ approximations of the manifold.

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